

What are the goals of this PhD research work?

Current Gap in AAM – What communication technologies like 5G, Wi-Fi and Satcom mean to AAM?

Ensures reliable data exchange between UAVs and ground stations for fail-safe operations.

Facilitates real-time navigation, surveillance, and control to enhance safety and efficiency.

How to ensure reliability?

Use a hybrid approach (multi-link communication) with 5G, Wi-Fi, and Satcom to enhance coverage.

Mesh technology will be a key factor in enabling the quick decision for conflict avoidance.

Prevents single points of failure and ensures continuous connectivity.

Is redundant transmission sufficient?

Integrate diverse communication channels (air-to-ground, air-to-air) for robustness.

Optimize bandwidth by transmitting computed data rather than raw sensor inputs.

Coded multiple communication network system between air, ground. And transmit computed data.

But the network technologies are not ideal in all the condition, how to handle this in different conditions while flying?

Adapt to real-world conditions through dynamic switching between networks.

Continuously assess network performance and refine communication algorithms.

Combined Hardware and software simulation for deterministic results

While network software simulations help in initial testing and validation, they are not sufficient for fully implementing and deploying a coded multi-link architecture in real-world BVLOS UAV operations. Here the network does not have idealized behavior. Also, onboard computation for multi-path decision is a key factor. More knowledge about network dynamics → better understanding of processing constraints and optimization of resource usage

Implementation strategies

Stages	Timeline	Equipment /technology	Setup	Independent outcome	Collaborative outcome
1.1	2025	Wi-Fi Mesh	Pixhawk to ground C2 communication	Control flight controller Network quality influence monitoring	Partial with T2
1.2	2025	Wi-Fi Mesh	Camera/Sensor to ground	Monitor resource allocation	T5, T8 (nav data)
1.3	2025	5G	Pixhawk to ground C2 communication	Control flight controller Network quality influence monitoring	Partial with T2
1.4	2025	Satcom	Pixhawk to ground C2 communication	Control flight controller	Partial with T2

				Network quality influence monitoring	
2.1	2026	Wi-Fi-Mesh	Air – to-Air collision avoidance	Send short messages – Broadcast vs unicast study	T6, T10
2.2	2026	5G + Satcom	Coded Multi-path architecture	Resource allocation, monitoring	T5, T8,
2.3	2026 - 2027	Ground multi-link management software testbed	Edge computing	Improve decision making, reliability model	T2, T5, T6, T8, T10

Note: each stages have internally different experiment setup (hardware /software in the loop)

Ambition: HW test with 5-6 drones - software – create docker container-based emulator to evaluate for 50 drones

Final test – Work the air-to-ground | Air-to-air | ground monitoring infrastructure together – can be a deliverable outcome from the project

Key outcomes of the test setup

1. Enable multi-link communication between drone and the ground station.
2. Enable communication quality monitoring system for multiple drones simultaneously and decision making based on external influences
3. Air-to -Air communication for conflict avoidance

Future cohort with this setup

1. Security planning
2. Enabling link quality prediction

Market study

Key criteria in mind while shortlisting candidates are as following:

1. Must support research – publish results – access to network quality parameters like signal strength, noise ratio etc., + size and weight of the hardware
2. Satcom – only LEO orbit satellite providers (latency restrictions)
3. Collaboration during experiments is a plus
4. Experience with communication on drones is a plus

Note: excluded 3DR Wi-Fi module – as its included in Holybro and premium modelbau set up – therefore cannot be used to mesh/broadcast algorithms.

Technology	Company	Outcome	Remarks by Shreeja
Wi-Fi	Tobufi – 8devices	OEM board with many features	3, but no housing – only touch with Sales team – No quotation received – Reach only via distributors
Wi-Fi	Meshmerize	With Mesh technology	1, 2, 3 also got to know RD team + access to some of mesh source code

			Experts at the chair – used – additional integration support acquired
5G	Raspberry pi 4B	Depend on no company , self building	Already placed for ordering – Do hardware in loop tests
5G	Sky-Drones AIRLink 5G Enterprise Set	Contacted sales team, no response with quotation	Enterprise kit – less access to network performance indicators are only on dashboard – Not useful for research
5G	Elsight Halo	Only ad emails – No point of contact	Other than information in their website. Found no collaborators. May be consider for future
5G	JOUAV 5G Networking UAV System	Must purchase with entire air unit and	No quotation till end of dec – Also, unsure if its suitable in EU spectrum
5G	Sierra Wireless AirLink XR60	few email exchange with the sales guy- let to no quotations	3
Satcom	SKYTRAC IMS Integrated Mission System & Satcom for UAV	No quotation	Trying to reach technical support – no avail
Satcom	Honeywell Verwave		Too big for the drones we have
Satcom +5G	Atmosphere GmbH	Integrated terminal - reasonable power consumption – access to network indicators	Worked with the terminal Familiar with the Iridium network and Atmosphere software Establish contact with the R&D team 1,2,3

Misc - *More depth into the research idea*

For Beyond Visual Line of Sight (BVLOS) UAV operations, robust and redundant communication is essential for reliability. The following technologies are suitable for BVLOS operations:

- Wi-Fi – Ideal for air-to-air tests, offering low latency and strong retransmission for minimal losses.
- 5G – Evaluates public network behavior, impact of link loss, and signal quality in flight.
- Satcom – Enhances reliability as a backup, with LEO satellites reducing latency.

Redundancy is the key - Instead of duplicating transmissions, a **coded multi-link system** can be implemented.

Data can be intelligently split across multiple links (Wi-Fi Mesh, 5G, Satcom) using network coding techniques.

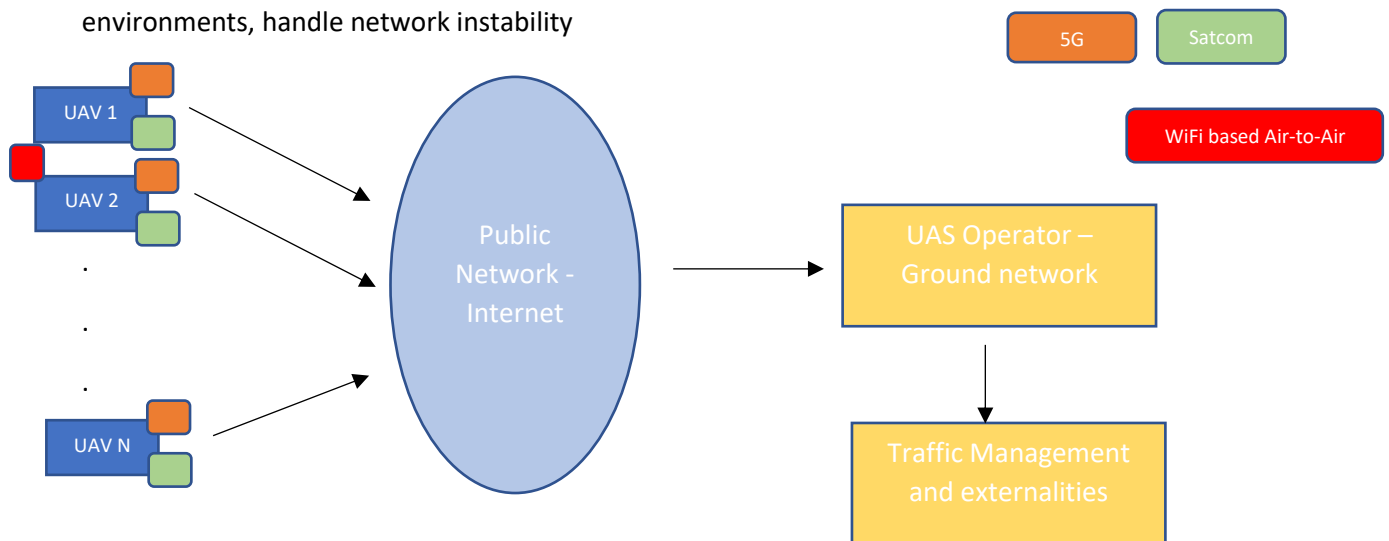
More spectrum doesn't always mean better communication opportunity.

The key is computational aspect to use the resources effectively

Proposal on this thesis: Beyond Simple Redundancy – Coded Multi-Link Architecture for UAV BVLOS (Adaptive Link Coding, Intelligent Error correction, Dynamic link selection)

Evaluation KPIs: **Bandwidth, latency and intelligent retransmission** - in line with ICAO, EASA, 3GPP and IEEE standards.

Additional parameters: link prioritization, switching, and bandwidth allocation in dynamic UAV environments, handle network instability



- The entities of the setup – will include the data type – control (MAVLink commands, video, sensor data). Dataflow is the key
- Based on the data type – implement traffic priorities, automated multipath decisions, cross-layer optimization

Summary of the outcomes

1. Proposed and evaluated Link Handle and link health system
 - UAVs continuously assess link latency, packet loss, and throughput, dynamically switching between links for stability.
 - Prioritizes traffic based on mission-critical vs. non-critical data,

- Enables dedicated network slices for UAVs, ensuring low-latency, high-reliability connectivity.
 - Processes UAV data closer to the edge, reducing backhaul congestion and improving real-time decision-making for navigation and object detection.
2. Open to Future – Network security in unlicensed spectrum and transformation impact on the industry